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A

B.Tech. Degree VI Semester Special Supplementary Examination November 2022

ME 19-205-0603 OPERATIONS MANAGEMENT (2019 Scheme)

Time: 3 Hours

Maximum Marks: 60

Course Outcome

On successful completion of the course, the students will be able to:

CO1: Understand different network techniques and forecasting methods.

CO2: Learn production planning and control as well as inventory control techniques.

CO3: Study the concepts of aggregate planning and scheduling.

CO4: Get the basic knowledge of plant location and lay out and material handling.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze,

L5 – Evaluate, L6 – Create

PO – Programme Outcome

PART A

(Answer *ALL* questions)

	(8 × 3 = 24)	Marks	BL	CO	PO
I. (a) Differentiate between PERT and CPM in network construction. List atleast one applications of each method.	3	3	L1	1	1
(b) What is crashing of network in project network? Explain with a suitable example.	3	3	L3	1	1
(c) What are the objectives of production planning and control?	3	3	L2	2	1
(d) Explain standardization and simplification in production planning.	3	3	L3	2	3
(e) What is the purpose of Gantt chart?	3	3	L1	3	1
(f) What is Aggregate planning?	3	3	L2	3	
(g) What are the different types of maintenance methods used in production industry?	3	3	L2	4	1
(h) What are the factors to be considered while selecting a plant location?	3	3	L3	4	2

PART B

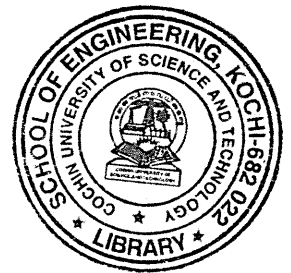
(4 × 12 = 48)

- II. A project consists of 4 activities. Their logical relationship and time taken is given along with crash time and cost details. If the indirect cost is ₹2000/- per week, find the optimal duration and optimal cost.

Activity	Predecessor	Normal Time		Crash Time	
		Time in days	Cost in ₹	Time in days	Cost in ₹
A	-	4	4000	2	12000
B	A	5	3000	2	7500
C	A	7	3600	5	6000
D	B	4	5000	2	10000
Total			15600		35500

OR

(P.T.O.)



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- III. Draw the network for the given project and find the probable project completion time.

Marks 12 BL L4 CO 1 PO 11

Activities		Time in Days		
i	j	to	tm	tp
1	2	5	12	17
1	3	8	10	13
1	4	9	11	12
2	3	5	8	9
2	5	9	11	13
4	6	14	18	22
3	7	21	25	30
6	7	8	13	17
6	8	14	17	21
7	8	6	9	12

- IV. (i) Explain the term product life cycle with the help of a diagram. 6 L3 2 1
(ii) What are the factors affecting product design? Explain. 6

OR

- V. (i) What is just in time and lean management. 6 L4 2 2
(ii) How a material requisition helps in procurement policy. 6
Explain the procedure adopted by a manufacturing industry for implementing procurement policy based on material requisition.

- VI. (i) What is sequencing? How it is applied in m machines - n jobs problems. 6 L4 3 2
(ii) Find the sequence of operation that minimise the total elapse time required to complete the job on machines M1 and M2 in the order M1M2. The processing time to complete the job in hours is given below. Also find the total elapsed time and the idle time for each machine 6

Job	A	B	C	D	E	F	G
M1	5	2	9	3	10	7	6
M2	2	6	7	8	4	4	3

OR

- VII. What is critical ratio in scheduling of jobs? Explain with a simple example of your choice. 12 L2 3 1

- VIII. (i) What is depreciation? How it affect the replacement policy? 12 L4 4 2
(ii) A firm is thinking of replacing a particular machine whose cost price is ₹25,200. The scrap value of the machine is ₹500/-. The maintenance costs are found to be as follows:

Year	1	2	3	4	5	6	7	8
Maintenance cost in ₹	420	700	1000	1300	1900	2500	3200	4000

Determine when the firm should get the machine replaced.

OR

- IX. Explain the different types of plant layout used in industry. Which layout is commonly used in coal industry? Why? 12 L4 4 2

Bloom's Taxonomy Levels

L1 = 5%, L2 = 17.5%, L3 = 27.5%, L4 = 50%.
